

**KONGU ENGINEERING COLLEGE**  
**(AUTONOMOUS)**

**PERUNDURAI, ERODE – 638 060**

**EXPERIMENT - 14**

**PROJECT TITLE**

**SMART INVENTORY MANAGEMENT SYSTEM  
FOR CONSTRUCTION SITES**

**SUBMITTED BY**

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# SMART INVENTORY MANAGEMENT SYSTEM FOR CONSTRUCTION SITES

## 1. Project review:

The **Smart Inventory Management System for Construction Sites** is designed to automatically monitor and manage the availability of construction materials. Construction sites require continuous monitoring of materials such as cement, sand, steel, bricks, and other equipment.

The proposed system uses an **ESP8266 NodeMCU** as the main controller along with an **ultrasonic sensor** to measure the material level or quantity in a storage container. The system continuously monitors the available inventory level.

When the material level falls below a predefined minimum level, the system activates a **buzzer and LED** to provide a low-stock warning. The ESP8266 can also use Wi-Fi connectivity to send inventory information for remote monitoring and management.

This system reduces manual inventory checking, prevents material shortages, and improves the efficiency of construction site management.

## 2. Problem Statement:

Construction sites require a large quantity of materials for continuous operation. Manual inventory monitoring can be time-consuming and may result in incorrect stock information.

A shortage of important construction materials may cause delays in construction work and increase project costs.

Therefore, an automatic system is required to continuously monitor the quantity of available materials and provide an alert when the inventory level becomes low.

The proposed system uses an **ultrasonic sensor, ESP8266 NodeMCU, buzzer, and LED** to automatically monitor inventory levels and provide low-stock alerts.

### **3. Proposed Solution:**

The proposed system consists of an **ESP8266-based Smart Inventory Management System**.

An ultrasonic sensor is placed near the storage container to measure the distance between the sensor and the material surface. The measured distance is used to determine the available material level.

The ESP8266 continuously processes the sensor readings and compares the inventory level with a predefined minimum level.

### **Proposed Operation:**

Start the system and initialize all components.

Measure the distance using the ultrasonic sensor.

Calculate the material level in the storage container.

Process the sensor data using ESP8266.

Compare the inventory level with the predefined minimum level.

Check whether the material quantity is sufficient.

If the inventory level is low, activate the warning system.

Turn ON the buzzer and LED.

Display “**LOW STOCK – MATERIAL REORDER REQUIRED**”.

Otherwise, display “**INVENTORY LEVEL NORMAL**”.

Repeat the inventory monitoring process continuously

### **4. System Architecture:**

The Smart Inventory Management System consists of the following sections:

#### **Input Section:**

Ultrasonic Sensor – Measures the distance between the sensor and the material surface.

The distance measurement is used to calculate the material level.

## Processing Section:

The **ESP8266 NodeMCU** acts as the central controller.

- Receives ultrasonic sensor data.
- Calculates the material level.
- Processes inventory information.
- Compares the inventory level with the minimum stock limit.
- Detects low-stock conditions.
- Controls the buzzer and LED.

## Alert Section

The ESP8266 provides Wi-Fi connectivity for future development.

The system can be extended for:

- Remote inventory monitoring.
- Cloud-based stock management.
- Automatic reorder notifications.
- Mobile alerts.
- Construction material tracking.
- Inventory data analysis.

## 5. Circuit Table:

Important Pin Mapping:

| ESP8266 NodeMCU Pin | Component                           |
|---------------------|-------------------------------------|
| D1                  | GPIO5 Ultrasonic Sensor → TRIG      |
| D2                  | GPIO4 Ultrasonic Sensor → ECHO      |
| D6                  | GPIO12 Buzzer → Positive Terminal   |
| D7                  | GPIO13 LED → Anode through resistor |
| 5V / 3.3V           | — Sensor Power                      |
| GND                 | — Common Ground                     |

## 6. Circuit Design:

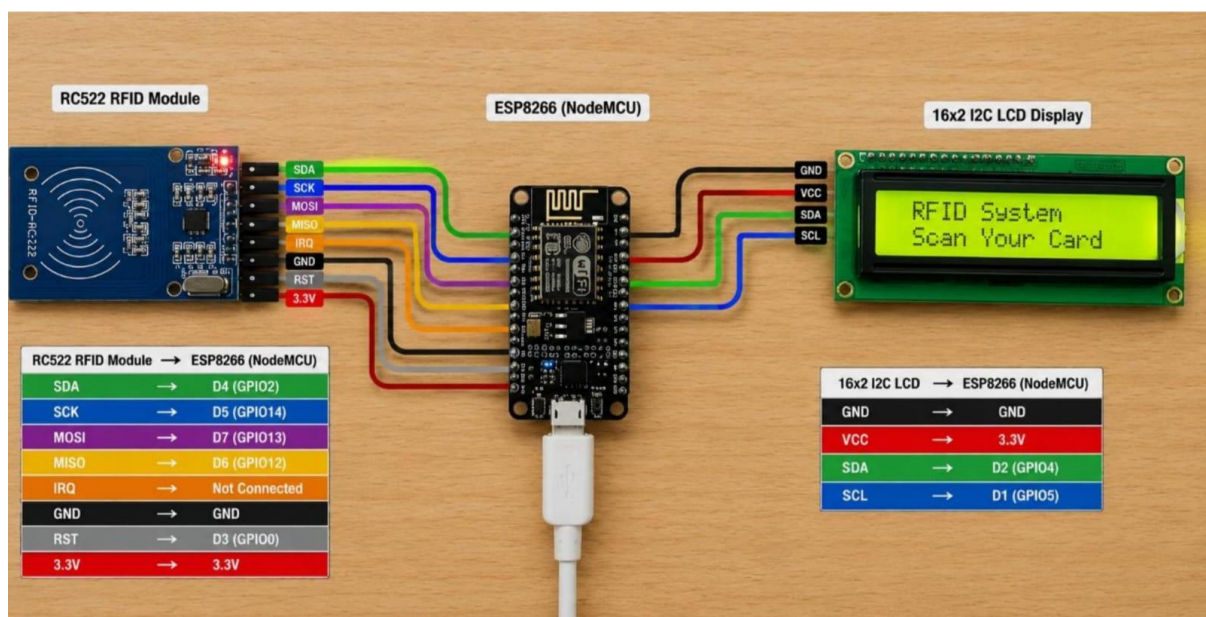
The circuit consists of an **ESP8266 NodeMCU**, **ultrasonic sensor**, **buzzer**, **LED**, **breadboard**, and **jumper wires**.

The **TRIG pin** of the ultrasonic sensor is connected to **D1 (GPIO5)**, and the **ECHO pin** is connected to **D2 (GPIO4)**.

The ultrasonic sensor measures the distance between the sensor and the surface of the stored material. The ESP8266 processes this distance and determines whether the inventory level is normal or low.

The buzzer is connected to **D6 (GPIO12)**, and the LED is connected to **D7 (GPIO13)** through a suitable resistor.

When the material level falls below the predefined limit, the ESP8266 activates the buzzer and LED to provide a low-stock warning.



## 7. Algorithm:

**Start** the system.

Initialize the **ESP8266 NodeMCU**, **ultrasonic sensor**, **buzzer**, and **LED**.

Trigger the ultrasonic sensor to measure the distance between the sensor and the material surface.

Calculate the **available material level** using the measured distance.

Process the inventory level using the ESP8266 NodeMCU.

Compare the material level with the predefined **minimum stock level**.

Check whether the inventory level is **below the minimum limit**.

If the stock level is low, turn **ON the buzzer and LED**.

Display “**LOW STOCK – MATERIAL REORDER REQUIRED**”; otherwise, keep the buzzer and LED **OFF** and display “**INVENTORY LEVEL NORMAL**”.

Wait for a short interval and **repeat the inventory monitoring process continuously**

## 8. Coding:

```
// =====  
// SMART INVENTORY MANAGEMENT SYSTEM  
// FOR CONSTRUCTION SITES  
// ESP8266 + ULTRASONIC SENSOR + BUZZER + LED  
// =====
```

```
#define TRIG_PIN 5
```

```
#define ECHO_PIN 4
```

```
#define BUZZER_PIN 12
```

```
#define LED_PIN 13
```

```
#define TANK_HEIGHT 30.0
```

```
#define LOW_STOCK_LEVEL 20.0
```

```
void setup()
```

```
{
```

```
  Serial.begin(9600);
```

```
pinMode(TRIG_PIN, OUTPUT);
```

```
pinMode(ECHO_PIN, INPUT);
```

```
pinMode(BUZZER_PIN, OUTPUT);
```

```
pinMode(LED_PIN, OUTPUT);
```

```
digitalWrite(BUZZER_PIN, LOW);
```

```
digitalWrite(LED_PIN, LOW);
```

```
Serial.println();
```

```
Serial.println("=====");
```

```
Serial.println(" SMART INVENTORY MANAGEMENT SYSTEM");
```

```
Serial.println("  FOR CONSTRUCTION SITES");
```

```
Serial.println("=====");
```

```
Serial.println("System Started");
```

```
}
```

```
void loop()
```

```
{
```

```
    long duration;
```

```
    float distance;
```

```
    float materialLevel;
```

```
    // Trigger ultrasonic sensor
```

```
    digitalWrite(TRIG_PIN, LOW);
```

```
    delayMicroseconds(2);
```

```
digitalWrite(TRIG_PIN, HIGH);
delayMicroseconds(10);

digitalWrite(TRIG_PIN, LOW);

// Read echo signal
duration = pulseIn(ECHO_PIN, HIGH);

// Calculate distance
distance = duration * 0.034 / 2;

// Calculate material level
materialLevel = TANK_HEIGHT - distance;

// Limit negative values
if (materialLevel < 0)
{
    materialLevel = 0;
}

Serial.println("-----");

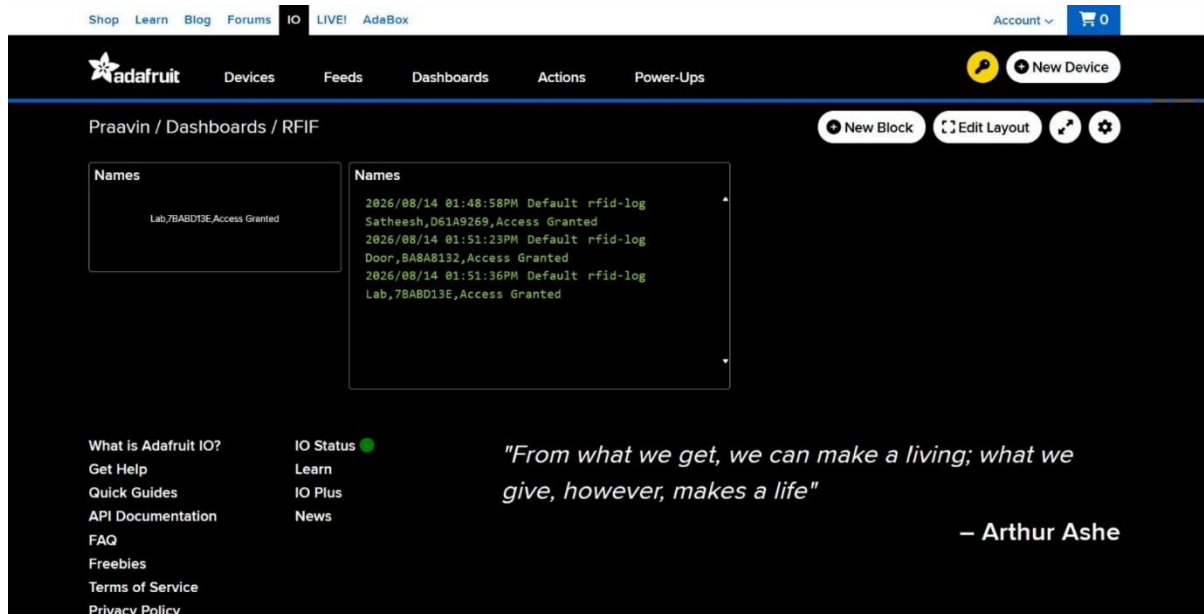
Serial.print("Distance from Material : ");
Serial.print(distance);
Serial.println(" cm");

Serial.print("Material Level : ");
```



```
Serial.print(materialLevel);  
Serial.println(" cm");  
  
// Check inventory level  
if (materialLevel < LOW_STOCK_LEVEL)  
{  
    digitalWrite(BUZZER_PIN, HIGH);  
    digitalWrite(LED_PIN, HIGH);  
  
    Serial.println("STATUS : LOW STOCK");  
    Serial.println("MATERIAL REORDER REQUIRED");  
}  
else  
{  
    digitalWrite(BUZZER_PIN, LOW);  
    digitalWrite(LED_PIN, LOW);  
  
    Serial.println("STATUS : INVENTORY NORMAL");  
    Serial.println("MATERIAL LEVEL SUFFICIENT");  
}  
  
delay(2000);  
}
```

## Coding Output:



## 9. System Working:

### Power ON the System:

The ESP8266 NodeMCU is powered ON.

### Initialize Components:

The ultrasonic sensor, buzzer, and LED are initialized.

### Measure Material Distance

- The ultrasonic sensor measures the distance between the sensor and the material surface.

### Calculate Material Level

The ESP8266 calculates the available material level using the measured distance.

### Check Inventory Status

The calculated material level is compared with the predefined minimum stock level.

## **Low Stock Condition**

If the material level is below the minimum limit:

- Buzzer → ON
- LED → ON
- Inventory Status → Low Stock
- Reorder Alert → Activated

## **Normal Stock Condition**

If sufficient material is available:

- Buzzer → OFF
- LED → OFF
- Inventory Status → Normal

## **Repeat the Process**

The system continuously monitors the available inventory level. OFF

## **Repeat the Process**

The system continuously receives and processes GPS data to monitor the vehicle location.

## **10. Prototype Implementation:**

The prototype is implemented using an **ESP8266 NodeMCU, ultrasonic sensor, buzzer, LED, breadboard, and jumper wires.**

The ultrasonic sensor is used to measure the distance between the sensor and the stored construction material. The ESP8266 calculates the material level and determines the inventory status.

## **Hardware Setup**

Connect the Ultrasonic Sensor TRIG pin → D1 (GPIO5).

Connect the Ultrasonic Sensor ECHO pin → D2 (GPIO4).

Connect the Buzzer → D6 (GPIO12).

Connect the LED → D7 (GPIO13) through a suitable resistor.

Connect the sensor power pins to a suitable power supply.

Connect all GND terminals to a common ground.

Upload the program using Arduino IDE.

Power ON the system.

Place the ultrasonic sensor above the material container.

Observe the measured distance and calculated material level.

### **Prototype Working:**

The ultrasonic sensor continuously measures the material distance.

The ESP8266 calculates the available material level.

The inventory level is compared with the predefined minimum limit.

If the material level is sufficient, the system indicates normal inventory.

If the material level becomes low, the buzzer and LED are activated.

The system provides an automatic warning for material reordering.

The process repeats continuously.

### **11. Bill of Materials:**

| S.No. | Component         | Specification                          | Quantity |
|-------|-------------------|----------------------------------------|----------|
| 1     | ESP8266           | NodeMCU Wi-Fi Development Board        | 1        |
| 2     | Ultrasonic Sensor | HC-SR04 or Equivalent                  | 1        |
| 3     | Buzzer            | 3.3V/5V Buzzer                         | 1        |
| 4     | LED               | Standard Indicator LED                 | 1        |
| 5     | Resistor          | Suitable LED Current Limiting Resistor | 1        |
| 6     | Breadboard        | Standard Prototype Board               | 1        |
| 7     | Jumper Wires      | Male-Male / Male-Female                | 1 Set    |

| S.No. | Component    | Specification             | Quantity |
|-------|--------------|---------------------------|----------|
| 8     | USB Cable    | Micro-USB                 | 1        |
| 9     | Power Supply | Suitable Regulated Supply | 1        |

## 12. Results & Performance Analysis:

The prototype successfully demonstrates the basic operation of a **Smart Inventory Management System for Construction Sites**.

- The ultrasonic sensor successfully measures the distance to the stored material.
- The ESP8266 processes the sensor readings and calculates the material level.
- The system continuously monitors the inventory level.
- Low-stock conditions are automatically detected.
- The buzzer provides an audible warning.
- The LED provides a visual indication.
- The system can be extended using IoT connectivity for remote inventory monitoring and automatic material reorder notifications.

## Results

The ultrasonic sensor successfully measures the material distance.

The ESP8266 successfully calculates the inventory level.

The system continuously monitors construction material availability.

Normal and low-stock conditions are automatically identified.

The buzzer provides a low-stock warning.

The LED provides a visual indication.

The system reduces the need for manual inventory checking.

The prototype demonstrates the basic operation of a **Smart Inventory Management System for Construction Sites**.

## **Performance Analysis:**

| <b>Parameter</b>           | <b>Expected Performance</b> |
|----------------------------|-----------------------------|
| Material Level Monitoring  | Continuous                  |
| Distance Measurement       | Automatic                   |
| Data Processing            | Real-time                   |
| Inventory Status Detection | Automatic                   |
| Low Stock Detection        | Automatic                   |
| Audible Alert              | Buzzer ON                   |
| Visual Alert               | LED ON                      |
| Normal Stock Condition     | Buzzer and LED OFF          |
| Remote Monitoring          | Can Be Extended Using IoT   |

